Planning, Learning and Intelligent Decision Making
Homework 4 - Reinforcement Learning

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Exercise 1.

(a)

For this homework, we consider an MDP described by $(\mathcal{X}, \mathcal{A}, \{P_a\}, c, \gamma)$, where $\mathcal{X} = \{x\}$ and the cost-to-go function of an arbitrary policy π verifies

$$J^\pi(x) = c_\pi(x) + \gamma J^\pi(x) = \frac{c_\pi(x)}{1-\gamma} \tag{1}$$

After some manipulation of the update from the TD(0) algorithm for this MDP, described in the homework, we get to

$$\dot{J} = \mathbb{E}\big[c_t + (\gamma - 1)J\big] \tag{2}$$

Let's prove that J^π is an equilibrium point of the o.d.e. (2). J^π is an equilibrium point of (2) if

$$\mathbb{E}_\pi\big[c_t + (\gamma - 1)J^\pi\big] = 0 \tag{3}$$

It's important to note that $\mathbb{E}\big[c_t\big]$ for a policy π is equal to $\mathbb{E}_\pi\big[c_\pi\big]$. We can now apply (1) to (2), taking into account that $\mathbb{E}_\pi\big[c_t\big] = \mathbb{E}_\pi\big[c_\pi\big]$:

$$\begin{aligned}
&\mathbb{E}\big[c_t + (\gamma - 1)J^\pi\big] = 0 \\
\Leftrightarrow\ &\mathbb{E}\big[c_\pi + (\gamma - 1)J^\pi\big] = 0 \\
\Leftrightarrow\ &\mathbb{E}\Big[c_\pi + (\gamma - 1)\Big(\frac{c_\pi}{1-\gamma}\Big)\Big] = 0 \\
\Leftrightarrow\ &\mathbb{E}\big[c_\pi(1-\gamma) + (\gamma - 1)c_\pi\big] = 0 \\
\Rightarrow\ &\mathbb{E}\big[0\big] = 0
\end{aligned}$$

Note: As in the homework statement to simplify, since there is only one state x, we dropped the explicit mention of the state.

(b)

We want to prove that along the trajectories of the o.d.e. (2) the energy

$$E_t = \frac{1}{2}(J^{(t)} - J^\pi)^2 \tag{4}$$

always decreases, i.e., that $\dot{E}_t \leq 0$.

$$\begin{aligned}
\frac{\partial E_t}{\partial t} &= \frac{\partial E_t}{\partial J(x_t)} \cdot \frac{\partial J(x_t)}{\partial t} \\
&= (J^{(t)} - J^\pi) \cdot \frac{\partial J(x_t)}{\partial t} \\
&= (J^{(t)} - J^\pi)\Big((\gamma - 1)(J^t - J^\pi)\Big) \\
&= (\gamma - 1)(J^t - J^\pi)^2
\end{aligned}$$

If $\gamma < 1$ the $\dot{E}_t < 0$.
Since,

$$\begin{aligned}
\frac{\partial J(x_t)}{\partial t} &= \frac{\partial J^t}{\partial t} - \frac{\partial J^\pi}{\partial t} \\
&= \frac{\partial(c(x_t) + (\gamma - 1)J(x_t))}{\partial t} - \frac{\partial(c(x_\pi) + (\gamma - 1)J^\pi)}{\partial t} \\
&= c_\pi + (\gamma - 1)J^t - c_\pi - (\gamma - 1)J^\pi \\
&= (\gamma - 1)(J^t - J^\pi)
\end{aligned}$$

(c)

The energy, $E_t = \frac{1}{2}(J^{(t)} - J^\pi)^2$, represents the square error relative to the policy π, we showed in (b) that along the trajectory of the o.d.e. (2) the energy is always decreasing, meaning that the error is iteratively decreasing at each update of the TD(0) algorithm.